Yunbei Zhang

yunbeizhang.ml@gmail.com +1 (408) 896-0102 Mountain View, CA, USA

yunbeizhang.github.io Scholar LinkedIn GitHub

Research Statement

I work on the **safety and security of frontier AI agents**, motivated by reducing catastrophic risks from advanced AI. As models gain autonomy, tools, third-party skills, and the ability to delegate and self-improve, I study how safety failures emerge at the system level, and I build **automated red teaming**, **agentic safety evaluations**, and **control mechanisms** that make these failures measurable, attributable, and repairable. I intend to pursue AI safety research full-time after my Ph.D.

Education

- **Tulane University** — *Aug 2022 – Aug 2027 (Expected)*
  Ph.D. in Computer Science — New Orleans, LA, USA
- **Osaka University** — *Apr 2018 – Mar 2022*
  B.S. in Mathematics (Award of Excellence and multiple merit-based scholarships) — Osaka, Japan
- **Peking University** — *Jul 2013 – May 2015*
  Undergraduate dropout (Withdrew after two years to pursue alternative academic opportunities) — Beijing, China

Research Experience

- **Waymo** — *Jun 2026 – Aug 2026*
  Research Scientist Intern – Generative Planning & Safety (Mentors: Chi-Pang Lam and Hengbo Ma) — Mountain View, CA, USA
  - Developed **RL-based post-training for flow-matching trajectory generation** in autonomous driving, using collision-aware rewards to improve **long-horizon planning and rare-event robustness** in safety-critical scenarios (**ICRA submission**).
- **Oak Ridge National Laboratory** — *Jan 2026 – May 2026*
  Graduate Research Intern – Foundation Models & Uncertainty Quantification (Mentor: Xiao Wang) — Oak Ridge, TN, USA
  - Developed **uncertainty quantification and efficient adaptation for foundation models** (e.g., weather forecasting), studying calibration and reliability under distribution shift at HPC scale (**two ECCV'26 papers, one NeurIPS'26 paper**).
- **Amazon** — *Sep 2025 – Dec 2025*
  Applied Scientist Intern – Multimodal LLM Safety & Red Teaming (Mentor: Chandan K. Reddy) — Seattle, WA, USA
  - Built **MM-Plan**, an RL-trained agentic planner for **automated multimodal red teaming** that elicits visual-reasoning-only safety violations in 8 frontier MLLMs (e.g., GPT-5, Claude 4.5 Sonnet, Gemini 2.5 Pro) with **2–5× higher attack success** than baselines, and released **VE-Safety**, a 440-instance benchmark spanning 15 safety categories (**ICLRW 2026 Best Short Paper Award**).
  - Designed **RL-based post-training and agentic evaluation** to harden MLLMs against adversarial visual–textual prompts.
- **KLA Corporation** — *May 2025 – Aug 2025*
  Machine Learning Research Intern – Efficient Vision Foundation Models (Mentor: Rajkumar Theagarajan) — Milpitas, CA, USA
  - **Distilled a large vision foundation model into a compact YOLO detector**, achieving an **18% improvement in detection mAP** with stronger gains on **out-of-distribution data** while retaining efficient inference.

Selected Publications (* equal contribution, full list on website)

1. **Y. Zhang**, J. Wang, S. Lyu, Y. Ge, K. Liang, Z. Jin, C. K. Reddy, J. Hamm.
   Safety Must Survive Self-Improvement: Why Failures Persist and How Agents Recover.
   Preprint, 2026. [paper] [project] [code]
2. **Y. Zhang**, S. Lyu, J. Wang, Y. Ge, J. Guo, J. Hamm, C. K. Reddy.
   Deny Without Disabling: Authorization-Paired Evaluation and Control for Multi-Agent Systems.
   NeurIPS 2026 Agents in the Wild Workshop (AIWILD). [paper] [project] [code]
3. **Y. Zhang**, Y. Ge, W. Xu, Y. Xu, J. Hamm, C. K. Reddy.
   Visual Exclusivity Attacks: Automatic Multimodal Red Teaming via Agentic Planning.
   ICLR 2026 Agents in the Wild Workshop (AIWILD). [paper] [project] [dataset] [code] **[Best Short Paper Award]** (1 of 237 submissions)
4. **Y. Zhang**, K. Mei, M. Liu, J. Wang, D. N. Metaxas, X. Wang, J. Hamm, Y. Ge.
   Agents in the Wild: Safety, Society, and the Illusion of Sociality on Moltbook.
   ICLR 2026 Agents in the Wild Workshop (AIWILD). [paper] [project] [code] **[Covered by Epsiloon]**
5. **Y. Zhang**, J. Wang, Y. Ge, W. Xu, J. Hamm, C. K. Reddy.
   Stop Comparing LLM Agents without Disclosing the Harness.
   ICML 2026 Agents in the Wild Workshop (AIWILD). [paper] [project] [code] **[Featured in State of AI Report 2026]**
6. B. Wei, **Y. Zhang**, J. Pan, K. Mei, X. Wang, J. Hamm, Z. Zhu, Y. Ge.
   ClawSafety: "Safe" LLMs, Unsafe Agents.
   Preprint, 2026. [paper] [project] [code]
7. H. Hao, D. Min, Z. Zhang, **Y. Zhang**, M. Xu, Y. Ge, L. Cheng.
   One Skill, Two Masters: Serving Users and Attackers through Skill Injection.
   Preprint, 2026. [paper] [code]
8. Y. Chung, **Y. Zhang**, N. Marrouche, J. Hamm.
   SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy of Synthetic Image Generation.
   USENIX Security, 2025. [paper]

9. **Y. Zhang**, S. Niu, C. Cai, F. Liu, J. Hamm.
Adapting in the Dark: Efficient and Stable Test-Time Adaptation for Black-Box Models.
NeurIPS, 2026. ICLR 2026 Test-Time Updates Workshop (TTU). [paper] [project] [code] **[Best Paper Award]** (1 of 186 submissions)

10. **Y. Zhang**, C. Cai, F. Liu, J. Hamm.
Prime Once, then Reprogram Locally: An Efficient Alternative to Black-Box Service Model Adaptation.
CVPR, 2026. [paper] [code] **[Highlight]** (3.6% selection rate)

11. **Y. Zhang**, A. Mehra, S. Niu, J. Hamm.
DPCore: Dynamic Prompt Coreset for Continual Test-Time Adaptation.
ICML, 2025. [paper] [code]

12. **Y. Zhang**, J. Wang, X. Xiao, J. Hamm, X. Wang.
UtoMe: Observation-Uncertainty-Guided Token Merging for Weather Foundation Models.
NeurIPS, 2026. IS&T Electronic Imaging 2026 (HPCI). [paper] [project] [code] **[Oral]**

13. **Y. Zhang***, Z. Jin*, Y. Liu, J. Wang, X. Zhang, et al.
LIBERO-MAX: Do Robot Policies Adapt When the World Changes?
NeurIPS 2026 Robotics World Modeling Workshop. [paper] [project] [code] **[Oral]**

14. X. Xiao*, C. Ma*, **Y. Zhang***, C. Liu, Z. Wang, Y. Li, L. Zhao, et al.
Not All Directions Matter: Towards Structured and Task-Aware Low-Rank Model Adaptation.
ACL (Main), 2026. [paper]

15. J. Wang*, **Y. Zhang***, Z. Ding, J. Hamm.
Doctor Approved: Generating Medically Accurate Skin Disease Images through AI-Expert Feedback.
NeurIPS, 2025. [paper] [code]

16. G. Chen, S. Niu, G. Li, **Y. Zhang**, S. Shan, C. Miao, J. Yang.
EVA-0: Test-Time Model Evolution with Only Two Forward Passes per Sample.
NeurIPS, 2026. [paper]

17. X. Hu, H. Ni, **Y. Zhang**, J. Hamm, Z. Li, Z. Ding.
Seeing Clearly, Reasoning Confidently: Plug-and-Play Remedies for Vision Language Model Blindness.
CVPR, 2026. [paper]

18. X. Xiao, C. Liu, C.-T. Liao, **Y. Zhang**, Q. Lan, Y. Wei, L. Zhao, et al.
Staying VIGILant: Mitigating Visual Laziness via Counterfactual Visual Alignment in MLLMs.
ECCV, 2026. [paper]

19. M. Liu, **Y. Zhang**, S. Liu, L. Wang, W. Zhang.
Wan-R1: Verifiable-Reinforcement Learning for Video Reasoning.
ECCV, 2026. [paper]

20. J. Wang, **Y. Zhang**, L. Zhao, X. Xiao, J. Hamm, X. Wang.
Less Tokens, Better Forecasts: Sparse Residual Routing for Efficient Weather Prediction.
ECCV, 2026. [paper]

21. X. Xiao, X. Li, **Y. Zhang**, C. Han, T. Liu, T. Wang, R. Jiang, J. Hamm, et al.
Layer-Specific Prompt Fusion Discovery via Differentiable Search in Vision Foundation Models.
ECCV, 2026. [paper]

22. A. Mehra, **Y. Zhang**, J. Hamm.
Understanding the Transferability of Representations via Task-Relatedness.
NeurIPS, 2024. [paper]

Media Coverage & Open-Source Safety Artifacts

- **Media:** Agent-harness study featured in the **State of AI Report 2026** and **AIDB** (author interview); AI agent societies in **Epsiloon** (French science magazine); LIBERO-MAX in **Synced**.
- **Open Source:** **MM-Plan** and **VE-Safety** (multimodal red teaming); **RSI-Safety** (self-improvement); **FlowReview** (multi-agent control); **ClawSafety**, **SkillSafety** (agent injection).

Selected Honors & Grants

Finalist, Anthropic AI Safety & Security Fellows (2026)
Best Paper Award, ICLR'26 Test-Time Updates Workshop
Lambda Research Grant, Lambda (2025)
Monbukagakusho Honors Scholarship, MEXT (2018, 2021)
School of Science Honor Scholarship, Osaka (2018–2021)
Best Short Paper Award, ICLR'26 Agents in the Wild Workshop
Tinker Research Grant, Thinking Machines Lab (2026)
Outstanding TA Award, Tulane University (2024)
Ichikawa International Scholarship (2021)
Award of Excellence, Osaka University (2019)

Technical Skills & Service

- **Programming & ML Frameworks:** **Python**, C++, Bash, LaTeX, NumPy, **PyTorch**, **JAX**, Hugging Face Transformers, TRL, PEFT, verl, DeepSpeed, vLLM, Weights & Biases.
- **Training Systems:** Multi-node training and evaluation of foundation models on **ORNL Frontier (AMD MI250X)** and **NVIDIA GPU clusters**, scaling workloads to **hundreds of compute nodes**; Slurm, AWS, and large-scale experiment management.
- **Reviewer:** NeurIPS 2025–2026 (**2025 Top Reviewer**), ICLR 2025–2026, ICML 2026 (**Gold Reviewer**), CVPR 2026, ECCV 2026, ACM MM 2025–2026, and TMLR.
- **Languages:** English (fluent), Japanese (fluent), Chinese (native).